Özyeğin University

Phys 551 Spring 2026

Homework 3 - The Importance Sampling Contest

Due on Mar 17th, 2026

You will bridge statistical theory and computational efficiency by overcoming the failure of Naive Monte Carlo in rare-event simulations. By requiring the use of Importance Sampling in a competitive format, you will be pushed beyond merely selecting a *valid* proposal distribution to engineering an *optimal* one. You must mathematically analyze the target integrand's unique geometry and creatively construct a closely matching sampling distribution to minimize computational waste, thereby mastering the core principles of high-performance variance reduction in extreme-value problems.

Problem 1:

Offshore Wind & Hurricane Risk – Points: 100

You are working as a risk analyst for an insurance syndicate underwriting a new offshore wind farm. You need to calculate the **Expected Annual Loss (EAL)** per turbine due to catastrophic wind events to price the insurance premium accurately.

Meteorological data dictates that the 10-minute sustained wind speed v (in meters per second) at the hub height follows a **Weibull distribution** (specifically, a Rayleigh distribution with a scale parameter $c = 10$):

$$p(v) = \frac{v}{50} e^{-v^2/100}, \quad \text{for } v \geq 0$$

The wind turbine is designed to withstand normal storms. Its braking system feathers the blades and stops the rotor for any wind speed above 25 m/s. However, the structural yield limit of the tower is $v_{yield} = 45$ m/s, roughly a Category 4 hurricane.

For any wind speed $v < 45$, the financial loss is \$0. For wind speeds $v \geq 45$, the tower experiences plastic deformation, and the blades risk shattering. Based on the Ramberg-Osgood plasticity relationship, the financial cost of the damage (say, in USD) scales non-linearly with the excess wind speed:

$$C(v) = \begin{cases} 0 & \text{if } v < 45 \\ 10^5 \cdot (v - 45)^3 & \text{if } v \geq 45 \end{cases}$$

Your Goal: Calculate the Expected Annual Loss (EAL):

$$\text{EAL} = \int_0^\infty C(v)p(v)dv = \int_{45}^\infty 10^5 (v-45)^3 \left(\frac{v}{50} e^{-v^2/100} \right) dv$$

The Problem with Naive Monte Carlo

The probability of a wind event exceeding 45 m/s is $e^{-45^2/100} = e^{-20.25} \approx 1.6 \times 10^{-9}$. If you attempt to evaluate this integral using Naive Monte Carlo by drawing from $p(v)$, your simulation will see almost no hits. It will return an expected loss of \$0 with a variance of \$0, suggesting you to give the insurance for free, which is obviously wrong. To price this multi-million dollar policy, you **must** use Importance Sampling to force the hurricane to happen, and then correctly un-bias the result.

The Contest

You must invent an analytical Proposal Distribution, $q(v)$, to perform Importance Sampling on this integral.

We define the theoretical variance of the Importance Sampling estimator for a single sample as:

$$V_q = \mathrm{Var}_q\left(C(V)\frac{p(V)}{q(V)}\right)$$

For a simulation with N samples, your empirical **Score** is calculated as the sample variance of your weighted evaluations:

$$\text{Score} = \text{SampleVariance}\left(C(v_i)\frac{p(v_i)}{q(v_i)}\right) \approx V_q$$

(Note: The standard error of your final EAL estimate is simply $\sqrt{Score/N}$).

Contest Grading Rules

60 points will be used to judge your overall methodology and code. The remaining 40 points are granted according to the contest.

Your initial contest grade is strictly determined by how your **Score** compares to the best score in the class. Choose $N = 100,000$ to ensure precision in your score benchmark.

Let V^* be the absolute lowest valid variance (Score) submitted by any student. Your grade ratio $R = \text{Score}/V^*$ dictates your **Initial Contest Grade**:

- $R \leq 1.25$: **40 points**
- $1.25 < R \leq 1.50$: **20 points**
- $1.50 < R \leq 2.00$: **10 points**
- $R > 2.00$: **0 points**

Anti-Collusion / Scaling Rule:
To maintain the competitive integrity of this assignment, if a highly optimized solution is widely shared, the grades will be dynamically curved.
Let μ_c be the average of the Initial Contest Grades across the entire class. If $\mu_c > 20$, all contest grades will be scaled down proportionally so that the new class average becomes exactly 20 points. Your final contest grade will be calculated as:

$$\text{Final Contest Grade} = \begin{cases} \text{Initial Contest Grade} & \text{if } \mu_c \leq 20 \\ \text{Initial Contest Grade} \times \left(\frac{20}{\mu_c}\right) & \text{if } \mu_c > 20 \end{cases}$$

Rules for $q(v)$

a) $q(v)$ **cannot** be evaluated using numerical integration or brute-force grid searches to invert its CDF. It must be built from standard analytical distributions (e.g., Exponentials, Gaussians, Gammas, Beta, or truncations/shifts of them) from which you can efficiently draw random samples using standard libraries like SciPy or NumPy.
b) $q(v)$ must be properly normalized ($\int q(v)dv = 1$).
c) You must explicitly write out the PDF of your chosen $q(v)$.

Deliverables - "The 60 points"

Please submit a single PDF containing the following:

a) **The Analytical Definition:** Write down the exact mathematical function for your chosen $q(v)$.
b) **The Justification:** Explain *why* you chose this function. *Hint: Where does the integrand $C(v)p(v)$ actually peak? Does it spike immediately at $v = 45$, or does it take time to build up? What shape best matches that "hump"?*
c) **The Sampling Strategy:** Briefly explain how your algorithm draws random samples from $q(v)$ (e.g., "I used the inverse transform method on a shifted exponential..." or "I generated a Gamma variable and shifted it to the right by 45...").
d) **The Code:** Provide your clean, commented Python/MATLAB/C++... code. It must use exactly $N = 100,000$ samples. Be sure to use `ddof=1` if using Python's variance function.
e) **The Results:**
   - Your estimated Expected Annual Loss ($\widehat{\text{EAL}}$) in dollars.
   - Your Contest Score (the sample variance of your N weighted evaluations).

A Warning

A common trap in extreme value Importance Sampling is to simply use a shifted exponential distribution: $q(v) = \lambda e^{-\lambda(v-45)}$. While this guarantees you will hit the rare event and get a finite answer, look closely at the shape of an exponential distribution: it has its absolute maximum exactly at $v = 45$.

Does our target integrand $C(v)p(v)$ have its maximum at $v = 45$? No, because $C(45) = 0$.

If your proposal distribution is peaking where the target function is zero, you are generating thousands of samples that contribute nothing to the integral, heavily inflating your variance. To win this contest, you must plot the integrand, find its peak, and choose a $q(v)$ that shares that exact same architecture.